ELECTRICAL RESISTIVITY TOMOGRAPHY INSTRUMENT

Sub-$250 open-source subsurface imaging · Wenner four-electrode array

BACKGROUND

NAE Grand Challenge
Engineer the Tools of Scientific Discovery

Subsurface imaging is inaccessible. Commercial ERT systems cost $5,000–$50,000 — beyond reach for communities that need them most.

Post-disaster housing in South and Southeast Asia is rebuilt on ground with no structural assessment. Families build on liquefied soil, voids, and unstable fill layers.

This instrument changes that.

DESIGN LINEAGE

- **Phase 1**
  FMCW Radar (HLK-LD2410C)
  RF cannot penetrate soil — abandoned
- **Phase 2**
  INA128 + IAO505S dual rail
  Part error → superseded by INA333
- **Phase 3**
  INA333 single-supply + MT3608
  Current design — production

APPLICATIONS

- Post-disaster housing assessment
- Archaeological site survey
- Agricultural soil moisture mapping
- Infrastructure void detection
- Geohazard and sinkhole screening
- Open-source educational platform

MISSION LINEAGE

NASA RIMFAX · ESA WISDOM
Both Mars instruments image the subsurface using electromagnetic methods. This project's pivot from RF radar to DC resistivity mirrors real geophysical instrument trade studies at mission design level.

COST COMPARISON

This instrument	$245
DIY ERT (published)	$500–2k
AGI SuperSting R1	$8,000
ABEM Terrameter LS2	$30,000

TSA ENGINEERING DESIGN · NATIONALS
Team 22803
Louisville, Kentucky

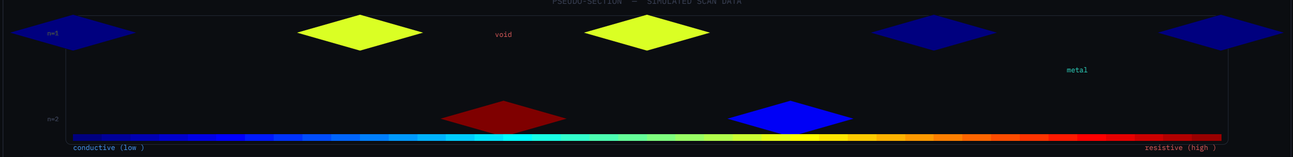
WENNER ARRAY GEOMETRY



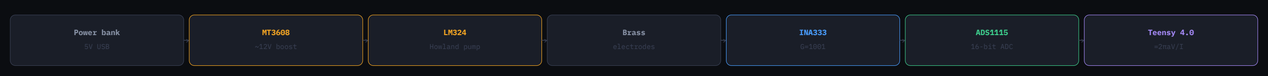
$$= 2\pi \cdot a \cdot (V_MN / I)$$

apparent resistivity (Ω·m) · a = electrode spacing · V_MN = measured voltage · I = injected current

PSEUDO-SECTION — SIMULATED SCAN DATA



SIGNAL CHAIN



KEY SPECIFICATIONS

Microcontroller	Teensy 4.0 · 3.3V · 600MHz Cortex-M7
ADC	2× ADS1115 16-bit · I²C · 0x48/0x49
Amplifier	INA333 G=1001 · 200pA bias · 100dB CMRR
Gain formula	G = 1 + (100k / R_G) ≈ 1 + (100,000/100)
Voltage reference	Ratiometric: (V_OUT − V_REF) / 1001
Current source	Howland pump · LM324 · TL431 · 2N3904
Compliance rail	MT3608 boost ~12V → LM324 V+ only
Electrode mux	4× CD74HC4067 · 2× MCP23017 · I²C control
Electrodes	8× brass rod ¼"×12" · 10cm Wenner spacing
Serial interface	t=self-test · l=live mode · g=full scan → CSV
Build cost	~$245 total · USB power bank portable

METHODS & RESULTS

BUILD SEQUENCE

0. **Power rails** — MT3608 → 12V · Teensy 3.3V verified
1. **IDC bring-up** — Uno R3 · 4 devices · 0x20/21/48/49
2. **Mux switching** — 4× CD74HC4067 · firmware self-test
3. **Current pump** — Howland · TL431 · sense resistor
4. **INA333 (soldered)** — Perfboard · ratiometric VREF read
5. **Teensy migration** — Swap pull-ups 5V→3.3V · re-verify
6. **First soil scan** — 8 electrodes · 'g' → CSV → viewer
7. **Collect data** — 3 soil states · poster screenshots

ENGINEERING CHALLENGES

- **Dual-rail amplifier** — INA333 single-supply eliminates isolated ±5V converter entirely
- **Dry-soil compliance** — MT3608 ~12V to LM324 only; 3.3V logic unaffected
- **5V pin tolerance** — Bring-up on Uno; swap pull-ups 5V→3.3V before Teensy
- **ADC resolution** — ADS1115 16-bit vs Teensy 12-bit; ~100× sensitivity gain
- **VREF pedestal drift** — Ratiometric: read VREF live; subtract in firmware
- **High-G on breadboard** — INA333 soldered to perfboard; parasitics eliminated

EXPECTED RESULTS

- **Void (plastic) detection** — High anomaly: +50–200% above background
- **Metal target detection** — Low anomaly: −30–80% below background
- **Dry sand baseline** — background 300–1000+ Ω·m
- **Damp soil baseline** — background 50–200 Ω·m
- **Ratiometric accuracy** — VREF drift cancelled; valid across temps
- **Current measurement** — True I via 10 sense; always valid

CONCLUSION

A sub-$250 ERT instrument that maps subsurface density anomalies using a Wenner four-electrode array, 16-bit ADC chain, and single-supply instrumentation amplifier.

The instrument makes a previously $5,000–$50,000 technique accessible to communities performing post-disaster ground safety assessment.

Open-source firmware, documented schematics, and a $245 BOM enable local replication worldwide.

FUTURE WORK

→ 16-electrode full pseudo-section
→ Inversion algorithm (apparent → true Ω)
→ Waterproof field enclosure (IP67)
→ Wireless data transmission (BLE)
→ Smartphone interface for field use

IEEE Sensors co-authorship · AI Scheduler
Bluegrass Youth Sustainability Council
Technology Student Association · Nationals